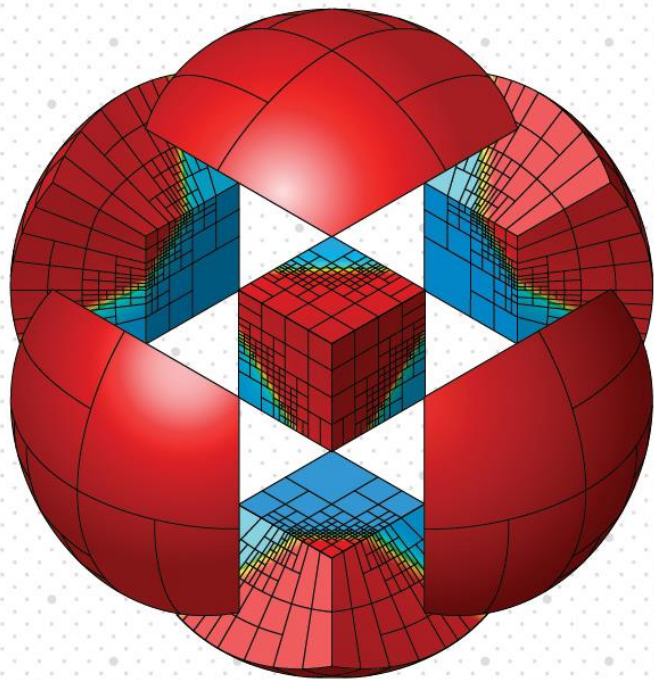




MFEM Community Workshop

Georgia Institute of **Technology**
Global Learning Center + Virtual
September 22-25, 2026



Welcome to the sixth annual MFEM Community Workshop!

Thank you to our organizing committee!



Holly Auten
LLNL



John Camier
LLNL



Brendan Keith
Brown University



Tzanio Kolev
LLNL



Will Pazner
Portland State University



Socratis Petrides
LLNL



Sohail Reddy
LLNL



Syun'ichi Shiraiwa
PPPL



Qi Tang
Georgia Tech

Thank You to the School of CSE Staff!

Thank you for making this workshop possible!



Sophie McGivern
Organization and registration



Della Phinisee
Finance and paperwork



Bryant Wine
Organization and
communications

Thank You to Our Sponsors

Gold Sponsor (\$6,000)



Silver Sponsors (\$4,000)



UK Atomic
Energy
Authority

Bronze Sponsor (\$2,000)



Georgia Tech College of Computing
School of Computational
Science and Engineering

Agenda

Full agenda: mfem.org/workshop/#agenda

	Wed, Sept 23	Thu, Sept 24	Fri, Sept 25
8am	Registration / Breakfast	Breakfast	Breakfast
	Welcome	Talk 9	2 Student Talks
9am	Talk 1	Talk 10	Talk 16
	Talk 2	Talk 11	Talk 17
10am	Coffee Break + Remote Photo	Coffee Break + Group Photo	Coffee Break + Vis Contest Awards
11am	Talk 3	Talk 12	Talk 18
	Talk 4	Talk 13	Talk 19
12pm	Lunch	Lunch	Lunch
1pm	Talk 5	Talk 14	Talk 20
	Talk 6	Talk 15	Talk 21
2pm	4 Student Talks	4 Student Talks	4 Student Talks
3pm	Coffee Break	Coffee Break	Coffee Break
4pm	Talk 7	Posters	Talk 22
	Talk 8		Talk 23

Thanks to Julian Andrej and Tzanio Kolev for yesterday's tutorial

Logistics

- **Schedule and links**

- Talks run 8:30 am to 5 pm all three days; **Zoom link and Slack invite were emailed**
- Join the MFEM Community Slack workspace for discussions during and after the workshop

- **Speakers**

- **In the room:** send your slides to Qi Tang (qtang@gatech.edu) and use the desktop; or use your own laptop via HDMI, connect to the Zoom and mute yourself
- **Remote:** share your screen on Zoom; please test audio and video before your session
- Talks are recorded by default and will be posted; tell us if you prefer not to be recorded
- Slides will be posted afterwards; **please send your slides to Qi Tang**

- **Meeting room**

- All workshop activities take place in this room (GLC Room 222)
- Wi-Fi: eduroam or GTvisitor

- **Group photo: Thursday 10:00 am, right after the morning session**

- **Dinner: nothing organized, check google map for options**

Request an Attendance Certificate



- Need proof of attendance for your institution?
- Request yours from Holly Auten on Slack (@HollyAuten) or at auten1@llnl.gov

New This Year: Poster Session

- **Thursday, 4:00–5:00 pm**
 - Georgia Tech GLC, in-person only
- **16 posters**
 - Set up during the Thursday afternoon coffee break
 - Presenters will get a gift (pins)



• Presenters

- Christine Tseng (Brown)
- Yiming Yu (Brown)
- Alexander de Magalhaes (GT)
- Brook Eyob (GT)
- David Winters (GT)
- Eddy Luo & Elliot Day (GT)
- Jackie Yao (GT)
- Meixi Li (GT)
- Peng Chen (GT)
- Rushan Zhang (GT)
- Tage Burnett (GT)
- Yanjie Tong (GT)
- Yuan Qiu (GT)
- Ilaria Fontana (Arizona)
- Leonardo Molinari (LLNL)
- Daniel Morales Bernal (UTSA)

New This Year: Challenge Coins

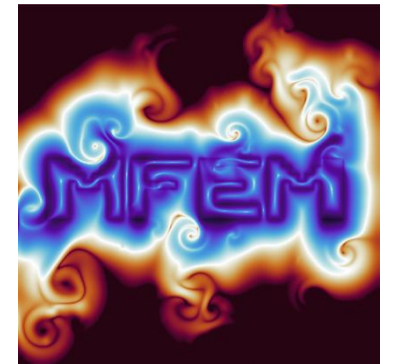
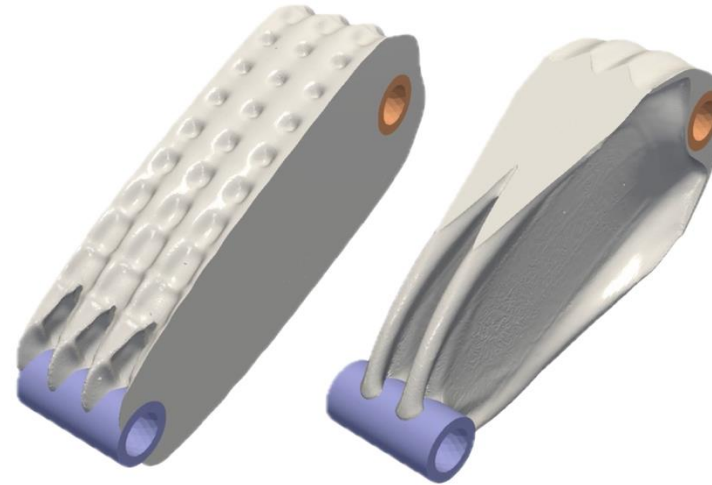
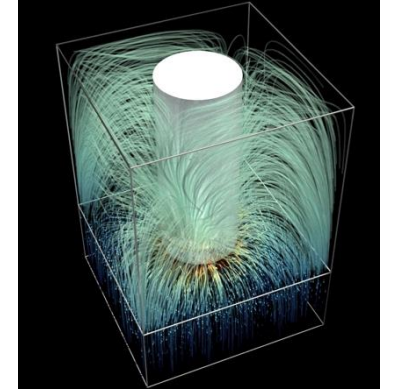
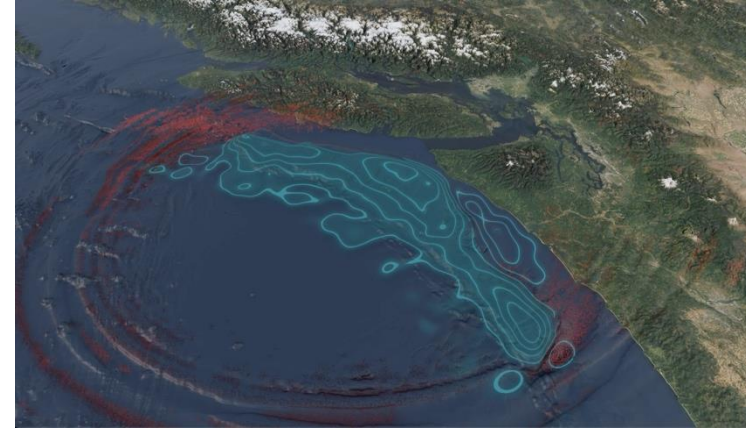
- All talk speakers and organizers receive an MFEM challenge coin
- Consider giving a talk next year!



Visualization Contest

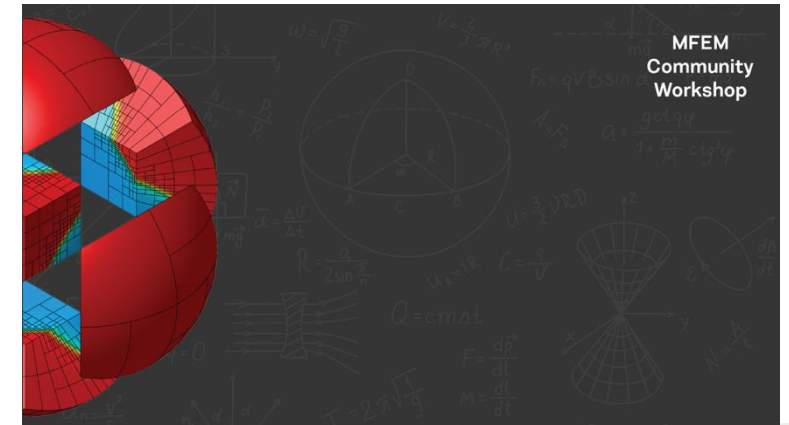
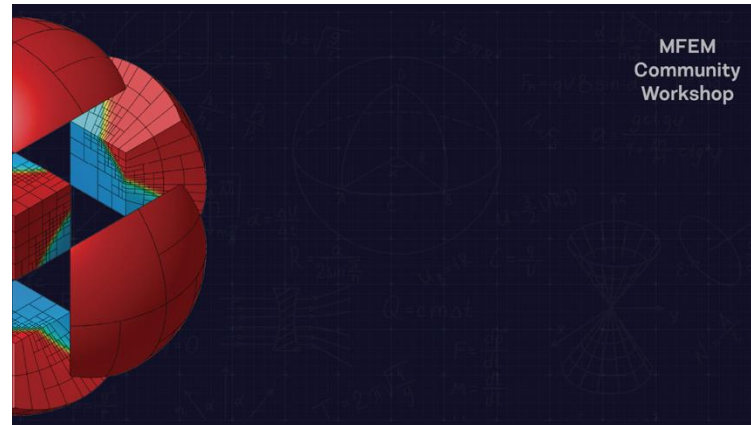
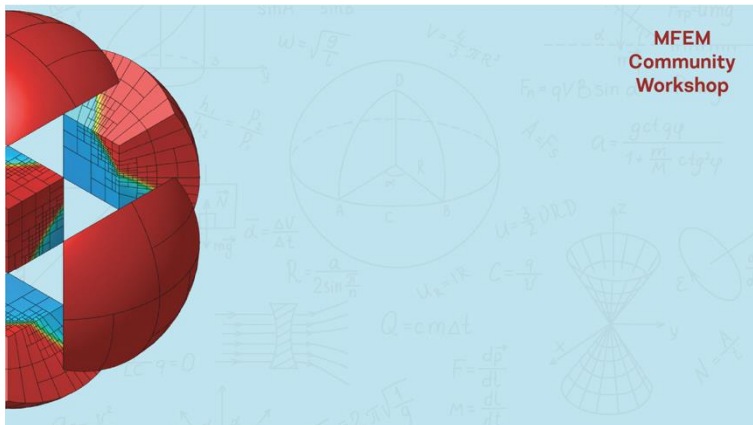
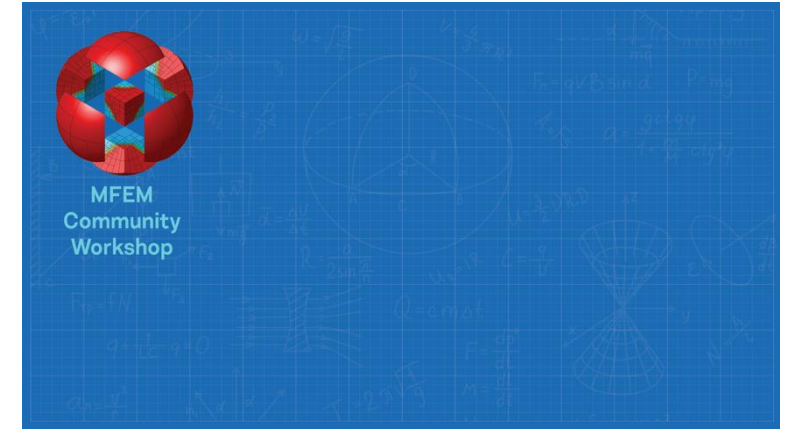
- Thank you for your submissions!
- Event organizers selected winners
- Winner announced Friday at the 10:00 am break
- The winners and runner-ups will receive an MFEM T-shirt and a challenge coin.
- Submissions will be featured at mfem.org/gallery

Some of the previous year winners



Download a Virtual Background

<https://mfem.org/workshop>



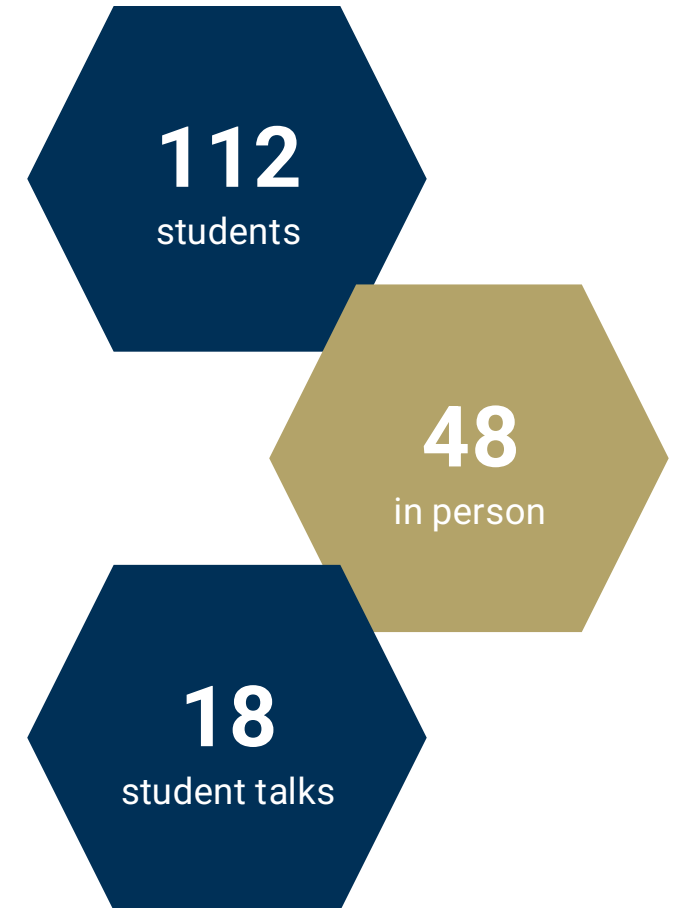
Workshop by the Numbers

- **304 registrants**
- **92 in person, 212 remote**
 - 41 Georgia Tech, 20 national labs, 3 UKAEA, 27 other universities, 1 industry (Coreform)
 - 5 traveled internationally (UK, Italy, Colombia)
- **~161 institutions**
 - ~110 universities, 26 labs and research institutes, 25 companies
- **37 countries**
- **37 talks and 16 posters**



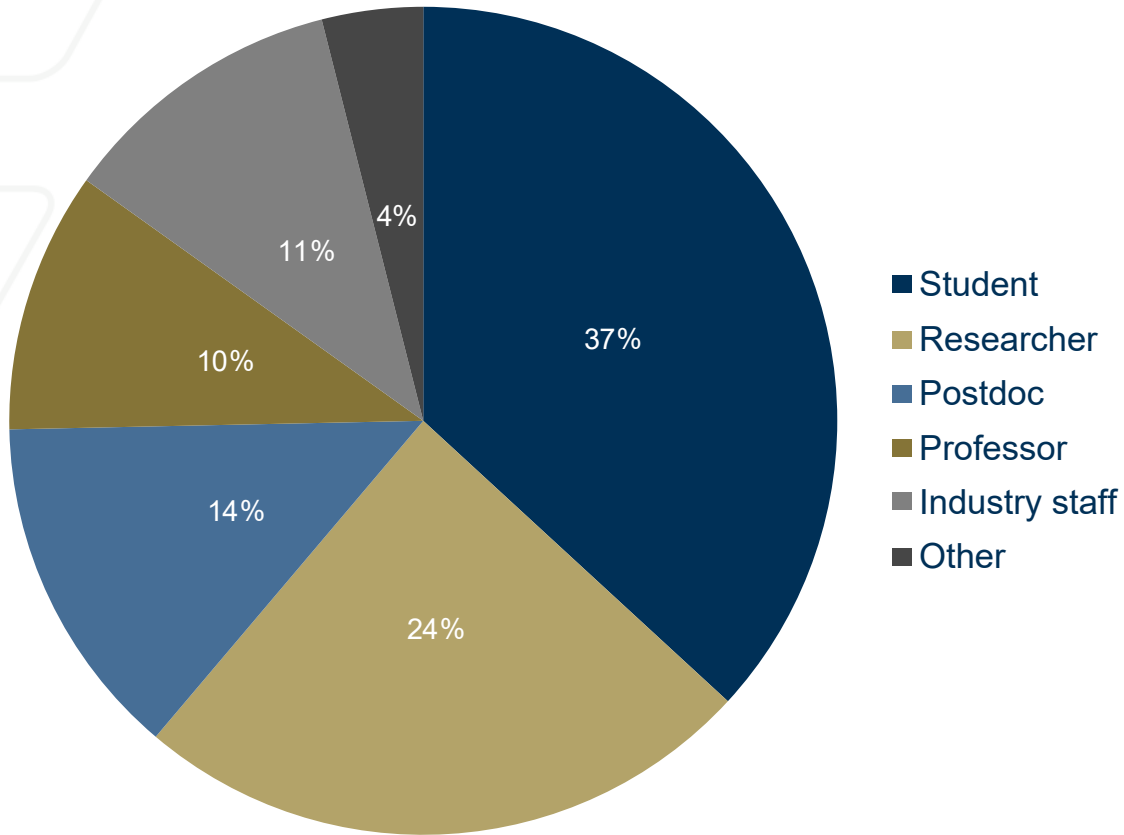
Students are a big part of the workshop

- **112 students registered, 48 in person**
- **4 student regular talks**
- **14 student lightning talks**
 - Each afternoon 2–3 pm (plus Friday 8:30 am)
- **Most student presenters are on site**
- **No travel funding this year, so thank you to the 17 students who traveled to Georgia Tech**

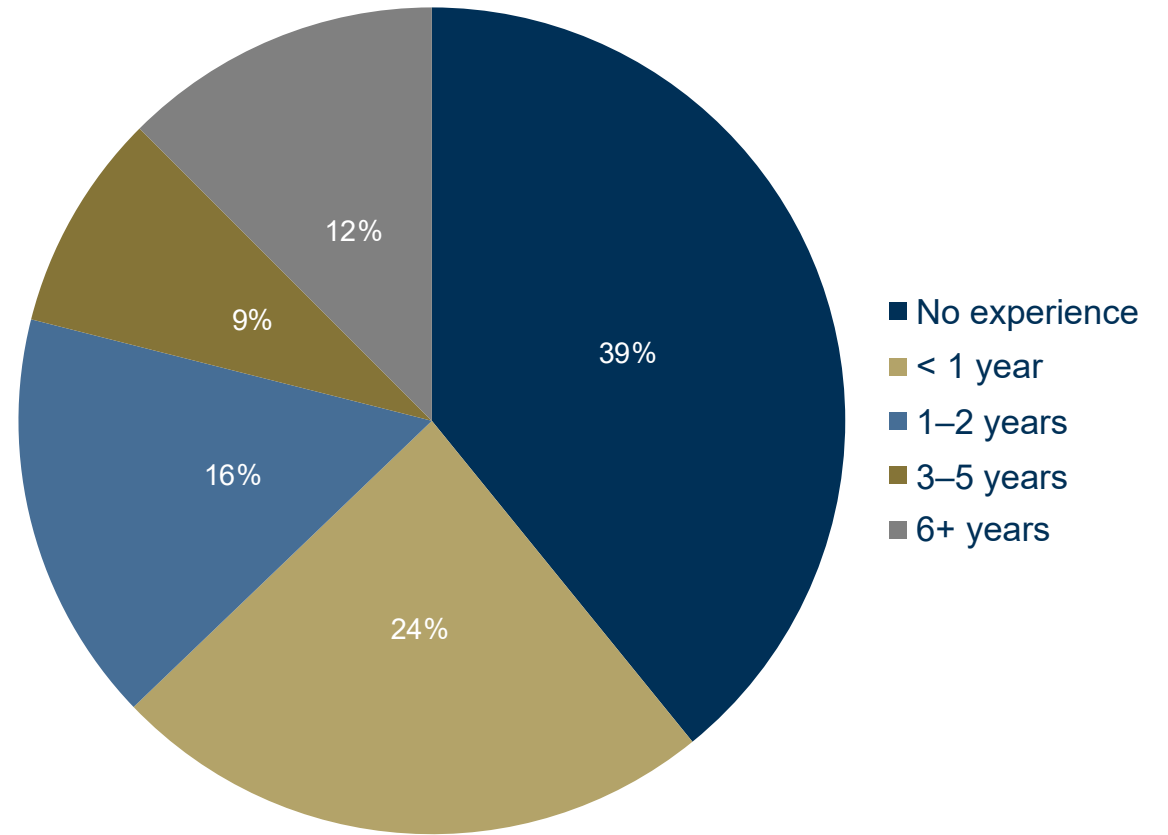


Who Is Here: Registration Survey

Position (304 responses)



Experience with MFEM (304 responses)



What You Told Us

- **Main Application Areas**

- Fluid flow (179)
- Structural mechanics (140)
- Data science (114)
- Electromagnetics (112)
- Design optimization (109)
- Fusion energy (75)
- Additive manufacturing (56)
- Subsurface flow (46)

- **Main Features of Interest**

- GPU acceleration (208)
- High-order finite elements (188)
- Adaptive mesh refinement (185)
- Linear solvers and preconditioners (168)
- Parallel scalability (156)
- Automatic differentiation (154)
- Mesh optimization (133)
- Variety of discretization methods (132)

What You Asked For

- **Help us review!**
 - Review is now the bottleneck, and AI-assisted PRs will add to it
 - Every experienced user can help: **become an MFEM reviewer**
- **Modernize for mfem-5.0**
 - Smart pointers, C++20
 - Cleaner class design: unify Bilinear/NonlinearForm DOF handling, less boilerplate
- **AI and scientific ML**
 - PyTorch / JAX coupling, neural operators
 - dFEM for mixed $H(\text{curl})$ / $H(\text{div})$ systems; more dFEM examples
 - Stable scripting interface so AI agents can drive setup, solve, and post-processing

From the survey

“Highest priority is improving timescales for review ... this is having a significant impact in encouraging adoption, both internally and with collaborators.”

— an MFEM user in industry/lab

Reviewing is the fastest way to shape MFEM and get your own PRs merged sooner.

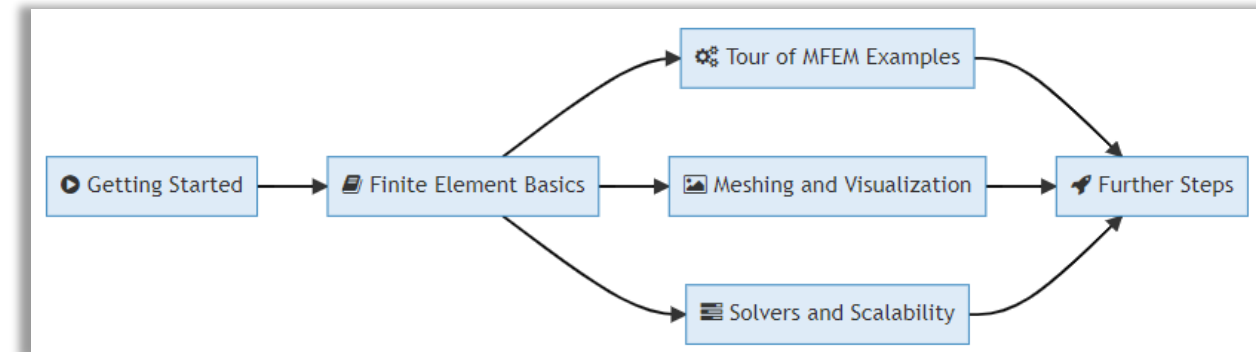
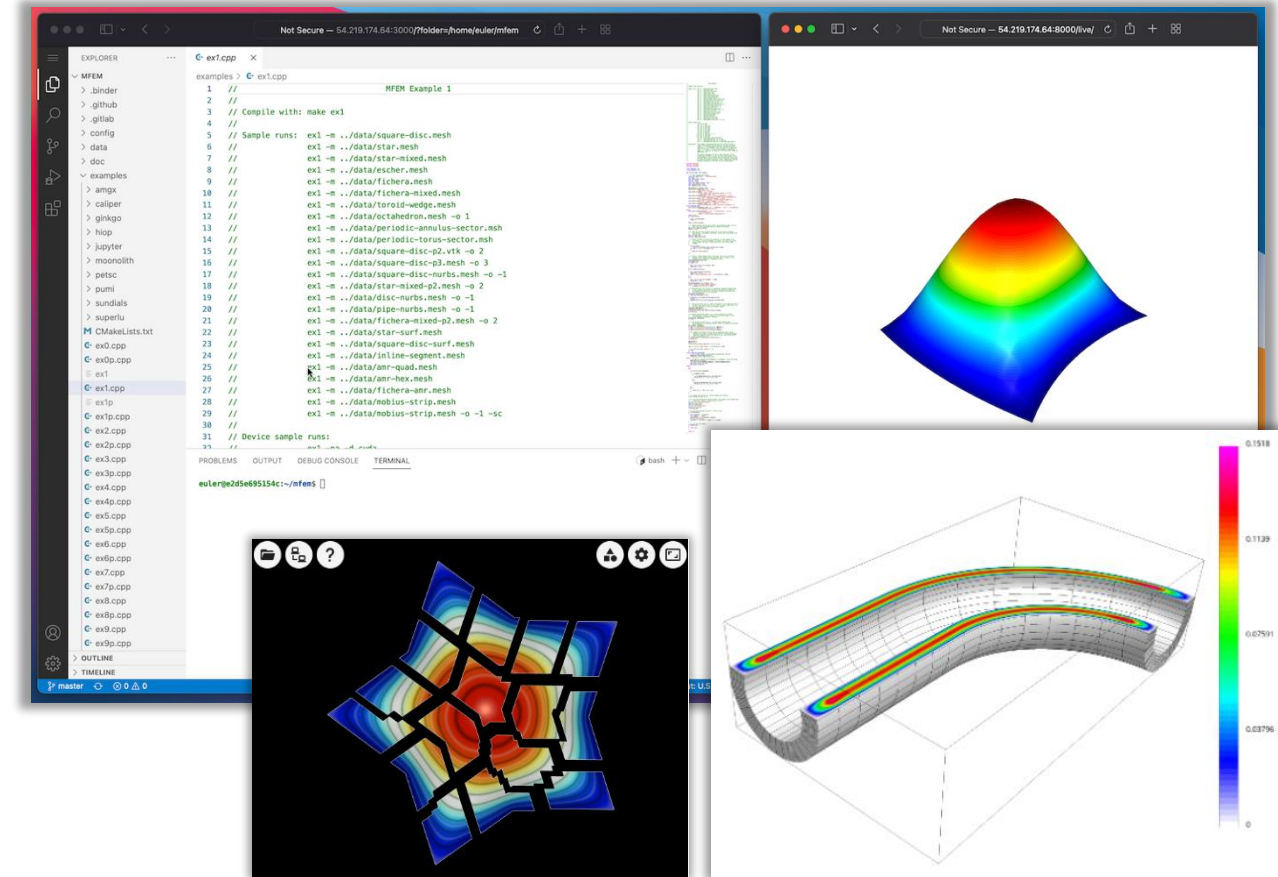
github.com/mfem/mfem/pulls

Let's Keep the Conversation Going

- Docs, examples, news, and more: mfem.org
- Github repos: github.com/mfem
- Stay in contact and ask for help:
 - mfem@llnl.gov
 - <https://github.com/mfem/mfem/issues>
- Join the MFEM Community Slack workspace and **#workshop-26** channel. All the announcements of the workshop will be posted there
- Become an MFEM reviewer!

Self-Paced MFEM Tutorial

- Available at mfem.org/tutorial
- No previous experience necessary
- Runs on an Amazon EC2 instance or a local Docker container
- Covers FEM basics, meshing, visualization, and scalable solvers
- Lots of examples, plus a video walkthrough (to be posted)



FEM@LLNL Seminar Series

- Roughly monthly on WebEx, 54 speakers so far
- Recordings on the MFEM YouTube playlist
- See the queue and sign up for notifications:
mfem.org/seminar
- Interested in giving a seminar? Write to mfem@llnl.gov

[MFEM](#) [Features](#) [Examples](#) [Documentation](#) [Community](#) [Gallery](#) [Download](#)

FEM@LLNL Seminar Series

The FEM@LLNL seminar series is focused on finite element research and applications talks of interest to the MFEM community. Videos will be added to a [YouTube playlist](#) as well as this site's [videos page](#).

✉ Sign-Up

Fill in [this form](#) to sign-up for future FEM@LLNL seminar announcements.



FEM@LLNL Seminar Series

by Inside Livermore Lab

Playlist • 33 videos • 1,339 views

▶ Play all

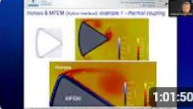
1



FEM@LLNL | Mesh Generation and Adaptation Using Green AI
Inside Livermore Lab • 5 views • 14 minutes ago

1:06:44

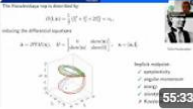
2



FEM@LLNL | New Avenues in Computational Fluid Dynamics
Inside Livermore Lab • 5 views • 16 minutes ago

1:01:50

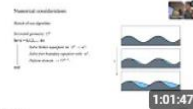
3



FEM@LLNL | Designing Conservative and Accurately Dissipative Numerical Integrators in Time
Inside Livermore Lab • 211 views • 2 weeks ago

55:33

4



FEM@LLNL | Numerical Solvers for Viscous Contact Problems in Glaciology
Inside Livermore Lab • 149 views • 4 months ago

1:01:47

The background of the slide is a faded, sepia-toned photograph of a large, multi-story historic building with many windows. In the foreground, a group of people is walking away from the camera on a path that leads towards the building. The overall tone is warm and nostalgic.

Enjoy the workshop!